Math 152 Lecture Notes: Week 7

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0.1 Review of Exponential Functions

Last week we introduced the natural exponential function

$$\exp : \mathbb{R} \to (0, \infty)$$

as the inverse of the natural logarithm. We saw that this function has the remarkable property that

$$\frac{\mathrm{d}}{\mathrm{d}x} \exp(x) = \exp(x)$$

and thus also

$$\int \exp(x) \mathrm{d}x = \exp(x) + C$$

We also saw that using exp and ln, we could define a general exponent a^r as $\exp(r \ln(a))$, where $a > 0$, $r \in \mathbb{R}$. Today will start with more consideration of the general exponential function a^x.

0.2 General Logarithm Functions

For $a, b > 0$, $a \neq 1$ we define $\log_a(b)$ as

$$\log_a(b) = \frac{\ln(b)}{\ln(a)}$$

This satisfies

$$\log_a(a^x) = \frac{\ln(\exp(\ln(a)x))}{\ln(a)} = \frac{\ln(a)x}{\ln(a)} = x$$

and also

$$a^{\log_a(x)} = \exp\left(\ln(a)\right)$$

That is to say, $\log_a(x)$ is the inverse to a^x.

Using the formula, we can differentiate $\log_a(x)$:

Proposition.

$$\frac{\mathrm{d}}{\mathrm{d}x} \log_a(x) = \frac{1}{x \ln(a)}$$

Because $\log_a(x)$ is a constant multiple of $\ln(x)$, we will get antiderivatives of general logarithm functions when we get an antiderivative for ln.

In the expression $\log_a(b)$, we can also hold b fixed and let a vary: the function $\log_x(b) = \ln(b)/\ln(x)$ makes sense where $x > 0$, $x \neq 1$, so on the intervals $(0, 1)$ and $(1, \infty)$.

Example 1. What is

$$\frac{\mathrm{d}}{\mathrm{d}x}\log_x(2)?$$

Writing

$$\log_x(2) = \frac{\ln(2)}{\ln(x)}$$

we have

$$\frac{\mathrm{d}}{\mathrm{d}x}\log_x(2) = \frac{-\ln(2)}{\ln(x)^2}\cdot\frac{1}{x} = -\frac{\ln(2)}{x\ln(x)^2}$$

History of the Exponential and the Number e

Our next topic will be an application of exponential functions. As context for this, I will say a bit about the history of the exponential function.

It is difficult to say when and by whom the exponential function was first introduced. This is because, as with the logarithm and many other mathematical concepts, there are many ways of getting at the exponential function, which were only later recognized to be related to each other. With this said, the general story, as far as I can tell, is that exponential notation a^x, including negative and rational x, developed through the 17th century, in parallel with the development of the logarithm. In the late 17th century, Jakob Bernoulli considered compound interest, where a debt grows in proportion to its current value. This led him to consider the number e as the limit $\lim_{n\to\infty}(1+1/n)^n$: the expression $(1+1/n)^n$ represents the result of compounding in n discrete intervals of length $1/n$. He also seems to have first considered exponential functions as inverses of logarithms. The derivatives of the exponential functions were first considered by Johann Bernouli, Jakob Bernouli's younger brother, in 1697. It was Leonhard Euler, in the early 18th century, who tied these threads together, considering the function e^x, which he recognized as the inverse to the natural logarithm, the limit $\lim_{n\to\infty}(1+x/n)^n$, and as the infinite series

$$\sum_n \frac{x^n}{n!} = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \dots$$

Infinite series like this are a major topic of Math 153.

1 Exponential Growth and Decay

The fact that the exponential function is its own derivative makes it a natural mathematical description of quantities that increase or decrease at a rate proportional to their current magnitude. The role of compound interest in the history of the exponential is one such situation. Here I will sketch out a few explicit examples of how to relate these descriptions to exponential functions.

The first example we consider is that of a population of bacteria in a Petri dish. The cells constantly divide, so the population grows at a rate proportional to its current rate. Letting $P(t)$ be the population at time t, we can write

$$P'(t) = k_{\text{growth}}\cdot P(t)$$

where $k_{\text{growth}} > 0$ is a constant, called the *growth constant*.

The second example we will consider is that of a radioactive isotope decaying. Each atom has the potential to decay, so if $m(t)$ is the mass of the isotope present at time t, $m(t)$ decreases in proportion to its current value:

$$m'(t) = k_{\text{decay}}\cdot m(t)$$

where $k_{\text{decay}} < 0$ is a constant, called the *decay constant*.

In both cases, we have a differential equation of the form

$$y'(t) = k \cdot y(t) \tag{$*$}$$

where k is a proportionality constant. We have seen that $y(t) = \exp(kt)$ satisfies this equation.

Proposition. *The only functions satisfying* $(*)$ *are those of the form* $y(t) = C\exp(kt)$ *for* C *a constant.*

Proof. Suppose y satisfies $(*)$ and consider

$$f(t) = y(t)/\exp(kt) = y(t)\exp(-kt)$$

By the product rule, we have $f'(t) = k\cdot y(t)\exp(-kt) - k\cdot y(t)\exp(-kt) = 0$ Therefore $f(t) = C$ is a constant, and $y(t) = C\exp(kt)$. □

Remark. The differential equation is equivalent to $\mathrm{D}\log(y) = k$. On the homework, you will show that the logarithmic derivative of a function determines the function up to a constant multiplicative factor.

Plugging in $t = 0$, we see that the constant C is $y(0)$. We can then conclude that the population of bacteria is described by

$$P(t) = P(0)\cdot\exp(k_{\text{growth}}t) = P(0)\cdot a_{\text{growth}}^t$$

where $a_{\text{growth}} = \exp(k_{\text{growth}}) > 1$. The mass of the decaying isotope is described by

$$m(t) = m(0)\cdot\exp(k_{\text{decay}}t) = m(0)\cdot a_{\text{decay}}^t$$

where $a_{\text{decay}} = \exp(k_{\text{decay}}) \in (0,1)$, in this case.

In both cases, the number a represents the proportion by which the quantity grows or shrinks during one unit of time:

$$y(t+1) = a\cdot y(t)$$

Another description of exponential growth or decay, which can be useful, instead asks for the period of time it takes for the quantity to change by a factor of 2. For exponential growth, we have the *doubling time* T_2. In every period T_2, the population doubles.

$$P(t+T_2) = 2\cdot P(t)$$

For decay, we have the *half life*, $T_{1/2}$ the time it takes for half of the material to disappear.

$$m(t+T_{1/2}) = \frac{m(t)}{2}$$

We can solve for these times in terms of the growth and decay constants, and vice versa. Using

$$P(t) = P(0)\exp(k_{\text{growth}}t)$$

the equation $P(T_2) = 2P(0)$ tells us that $\exp(k_{\text{growth}}T_2) = 2$, or in other words, $k_{\text{growth}}T_2 = \ln(2)$. The larger the growth constant, the shorter the doubling time. Similarly, using

$$m(T_{1/2}) = \frac{m(0)}{2}$$

we obtain $\exp(k_{\text{decay}}T_{1/2}) = 1/2$, or

$$k_{\text{decay}}T_{1/2} = \ln(1/2) = -\ln(2)$$

Keep in mind that k_{decay} is negative, so $T_{1/2}$ is still positive. The more negative k_{decay} is, the shorter the half life.

Example 2. Suppose a population of bacteria starts with 1000 individuals and doubles every 3 hours. What is its growth constant, and what is the population at time t?

Using the equation $k_{\text{growth}}T_2 = \ln(2)$, we have $k_{\text{growth}} = \frac{\ln(2)}{3}$. This means that

$$P(t) = P(0)\exp(k_{\text{growth}}t) = 1000\cdot\exp\left(\frac{\ln(2)}{3}t\right) = 1000\cdot 2^{t/3}$$

Digression: Radiocarbon Dating

This description of the decay of an isotope can be used to determine the age of samples containing organic material. Carbon occurs naturally in two stable isotopes and an unstable isotope, Carbon 14, which decays over time. New Carbon 14 forms when cosmic rays interact with gases in the atmosphere, so in the atmosphere, the proportions of these isotopes are stable over time.[1] When a plant is alive, it exchanges gas with the atmosphere, and its body has isotope ratios close to those of the atmosphere. When it dies, the isotope ratios change as the Carbon 14 decays. By measuring the change in ratio from the atmospheric isotope ratio and using our mathematical description of radioactive decay, we can calculate the age of a sample of plant matter.

2 Inverse Trigonometric Functions in Integration

Consider the accumulation function

$$F(y) = \int_0^y \frac{1}{\sqrt{1-t^2}} \mathrm{d}t$$

This makes sense on the interval $(-1, 1)$. If we plug $\sin(x)$ into this function, we can make the following observations:

$$\frac{\mathrm{d}}{\mathrm{d}x} F(\sin(x)) = \frac{\cos(x)}{\sqrt{1-(\sin(x))^2}}$$

using the fundamental theorem of calculus and the chain rule. When $x \in (-\pi/, \pi/2)$, $F(\sin(x))$ is well defined and this comes out to 1. Now $F(\sin(0)) = F(0) = 0$, so this tells us that $F(\sin(x)) = x$ when $x \in (-\pi/2, \pi/2)$. The accumulation function here appears to be an inverse of the sine function.

2.1 Inverse Sine

[Draw graph of sine function.]

The function $\sin : \mathbb{R} \to \mathbb{R}$ is not invertible: it is not injective, as $\sin(0) = \sin(\pi) = \sin(2\pi) = \ldots$, and it is not surjective, as it only takes values between -1 and 1. If we restrict the sine function to suitable intervals, we can find an inverse to it, which we can then use to express integrals like the one considered above.

Examining the graph of the sine function, we see that on the interval $[-\pi/2, \pi/2]$, it increases from -1 to 1. This can also be understood using the unit circle: [draw picture, with labeled point $(\cos(x), \sin(x))$] As the angle increases from $-\pi/2$ to $\pi/2$, the y-coordinate of the point increases from -1 to 1. Therefore, if we view sin as a function $\sin : [-\pi/2, \pi/2] \to [-1, 1]$, it has an inverse

$$\sin^{-1} : [-1, 1] \to [-\pi/2, \pi/2]$$

[Draw graph.] On the graph, we observe that $\sin^{-1}$ is an odd function. This is true because sin itself is odd.

Example 3. We have $\sin^{-1}(\sqrt{3}/2) = \pi/3$, $\sin^{-1}(1/2) = \pi/6$, $\sin^{-1}(-1/\sqrt{2}) = -\pi/4$. [Label points on unit circle.] Bear in mind that $\sin^{-1}$ only inverts the sine function on the interval $[-\pi/2, \pi/2]$. For example, $\sin^{-1}(\sin(5\pi/6)) = \sin^{-1}(-1/2) = -\pi/6$. [label on unit circle.]

Using the general formula for the derivative of an inverse function and the fact that

$$\frac{\mathrm{d}}{\mathrm{d}t} \sin(t) = \cos(t)$$

we have

$$\frac{\mathrm{d}}{\mathrm{d}x} \sin^{-1}(x) = \frac{1}{\cos(\sin^{-1}(x))}$$

[1] This is not quite true, the proportion of Carbon 14 in the atmosphere has varied throughout earth's history, and these variations require corrections to the carbon dating technique. In the past 200 years, large scale burning of fossil fuels and atmospheric nuclear testing have changed the isotope ratios by a significant amount.

where it exists. On the interval where $\sin^{-1}$ is defined, this is equivalent to an algebraic function. When $t \in [-\pi/2, \pi/2]$, $\cos(t) \geq 0$, so we have $\cos(t) = \sqrt{1 - \sin(t)^2}$ and thus, taking $t = \sin^{-1}(x)$,

$$\frac{\mathrm{d}}{\mathrm{d}x} \sin^{-1}(x) = \frac{1}{\cos(\sin^{-1}(x))} = \frac{1}{\sqrt{1-x^2}}$$

as we saw previously. Notice that this is undefined at $x = \pm 1$. The inverse sine function is only differentiable in the open interval $(-\pi/2, \pi/2)$, it has a vertical tangent line at the endpoints. [Refer to graph.]

Example 4. Where is $f(x) = \sin^{-1}(2x^2)$ defined? What is its derivative?

$\sin^{-1}$ is defined on $[-1, 1]$, and $2x^2$ lies in this interval when $|x| \leq 1/\sqrt{2}$, so f is defined on $[-1/\sqrt{2}, 1/\sqrt{2}]$. Using the chain rule, its derivative is

$$f'(x) = \frac{4x}{\sqrt{1-4x^4}}$$

We can take the derivative formula above and view it as an indefinite integral:

$$\int \frac{1}{\sqrt{1-x^2}} \mathrm{d}x = \sin^{-1}(x) + C$$

This antiderivative is well defined on the interval $(-1, 1)$ where the function in the integral is continuous.

Example 5. Consider the integral

$$\int_{-1/2}^{1/2} \frac{1}{\sqrt{1-x^2}} \mathrm{d}x$$

Using symmetry, this is

$$2 \int_{0}^{1/2} \frac{1}{\sqrt{1-x^2}} \mathrm{d}x = 2 \left[\sin^{-1}(x)\right]\Big|_0^{1/2} = 2 \cdot (\pi/6 - 0) = \pi/3$$

Using u-substitution, we can extend this formula to find integrals of more general algebraic functions of a similar form. We start by considering a function of the form

$$\int \frac{1}{\sqrt{a^2 - x^2}} \mathrm{d}x$$

Using the substitution $u = x/a$, we have $\mathrm{d}x = a\mathrm{d}u$, and $x^2 = a^2u^2$, so the integral can be rewritten as

$$\int \frac{a}{\sqrt{a^2 - a^2u^2}} \mathrm{d}u$$

assuning that $a > 0$, we can factor it out of the square root and cancel:

$$= \int \frac{a}{a\sqrt{1-u^2}} \mathrm{d}u = \int \frac{1}{\sqrt{1-u^2}} \mathrm{d}u = \sin^{-1}(u) + C = \sin^{-1}(x/a) + C$$

Example 6. Consider the integral

$$\int_{-1}^{0} \frac{1}{\sqrt{2-x^2}} \mathrm{d}x$$

Using the substitution $u = x/\sqrt{2}$ as above, this is

$$\left[\sin^{-1}(x/\sqrt{2})\right]\Big|_{-1}^{0} = 0 - \sin^{-1}(-1/\sqrt{2}) = -(-\pi/4) = \pi/4$$

2.2 Inverse Tangent

The tangent function

$$\tan(x) = \sin(x)/\cos(x)$$

is periodic with period π, and defined on intervals of the form $(n\pi - \pi/2, n\pi + \pi/2)$. [Draw graph]. On each such interval, it is increasing and ranges from $-\infty$ to ∞. This can be understood in terms of the unit circle. [Draw picture] As the angle ranges from $-\pi/2$ to $\pi/2$, the slope of the line increases from $-\infty$ to ∞.

A consequence of this is that if we view the tangent as a function $\tan : (-\pi/2, \pi/2) \to \mathbb{R}$, it is injective and surjective. We therefore have a well defined inverse $\tan^{-1} : \mathbb{R} \to (-\pi/2, \pi/2)$. [Draw graph]. This is an odd increasing function. It is a good example of a function $\mathbb{R} \to \mathbb{R}$ which is injective but not surjective.

Example 7. $\tan^{-1}(1) = \pi/4$, $\tan^{-1}(-\sqrt{3}) = -\pi/3$, $\tan^{-1}(\tan(7\pi/2)) = \tan^{-1}(1) = -\pi/4$. [Label points on graph and unit circle.]

As with the inverse sine function, we can use trigonometric identities to express the derivative of this function as an algebraic function. First of all, using $\tan(x) = \sin(x)/\cos(x)$ and the quotient rule, we have

$$\frac{\mathrm{d}}{\mathrm{d}t}\tan(t) = \frac{\mathrm{d}}{\mathrm{d}t}\frac{\sin(t)}{\cos(t)} = \frac{\cos(t)^2 + \sin(t)^2}{\cos(t)^2} = \frac{1}{\cos(t)^2} = \sec(t)^2$$

so using the formula for the derivative of an inverse function, we have

$$\frac{\mathrm{d}}{\mathrm{d}x}\tan^{-1}(x) = \frac{1}{\sec(\tan^{-1}(x))^2}$$

To write this as an algebraic function, we take the identity

$$\sin(t)^2 + \cos(t)^2 = 1$$

and divide through by $\cos(x)^2$ to obtain

$$\tan(t)^2 + 1 = \sec(t)^2$$

Taking $t = \tan^{-1}(x)$, we obtain $\sec(\tan^{-1}(x))^2 + 1 + x^2$, and therefore

$$\frac{\mathrm{d}}{\mathrm{d}x}\tan^{-1}(x) = \frac{1}{1+x^2}$$

Equivalently, we have

$$\int \frac{1}{1+x^2}\mathrm{d}x = \tan^{-1}(x) + C$$

This is valid everywhere. Both the function in the integral and the inverse tangent function are defined and continuous everywhere.

Example 8.

$$\int_{-1}^{1} \frac{1}{1+x^2}\mathrm{d}x = \tan^{-1}(1) - \tan^{-1}(-1) = \pi/4 - (-\pi/4) = \pi/2$$

As with the inverse sine function, we can extend the reach of this integral formula using substitution. Consider the integral

$$\int \frac{1}{a^2+x^2}\mathrm{d}x$$

Using the substitution $u = x/a$, this becomes

$$\int \frac{a}{a^2+a^2u^2}\mathrm{d}u = \frac{1}{a}\int \frac{1}{1+u^2}\mathrm{d}u = \frac{\tan^{-1}(u)}{a} + C = \frac{\tan^{-1}(x/a)}{a} + C$$

Example 9. Consider the integral

$$\int_0^3 \frac{2+2x}{9+x^2}\mathrm{d}x$$

We can break this up as a sum

$$\int_0^3 \frac{2}{9+x^2}\mathrm{d}x + \int_0^3 \frac{2x}{9+x^2}\mathrm{d}x$$

The first integral can be computed using the formula above:

$$\int_0^3 \frac{2}{9+x^2}\mathrm{d}x = \left[\frac{2\tan^{-1}(x/3)}{3}\right]\Bigg|_0^3 = \frac{2}{3}\tan^{-1}(1) = \pi/6$$

The second integral can be computed using the substitution $u = 9 + x^2$

$$\int_0^3 \frac{2x}{9+x^2}\mathrm{d}x = \int_9^{18} \frac{1}{u}\mathrm{d}u = [\ln(u)]|_9^{18} = \ln(18) - \ln(9) = \ln(18/9) = \ln(2)$$

Putting these together, we have

$$\int_0^3 \frac{2+2x}{9+x^2}\mathrm{d}x = \pi/6 + \ln(2)$$

3 Integration by Parts

The method of integration by substitution is based on the chain rule for differentiation. Applying the chain rule backwards, we can find antiderivatives and thus compute integrals using the fundamental theorem of calculus. The next technique we will learn, integration by parts, is derived from the product rule in the same way.

Consider a product of functions $u(x)v(x)$. Differentiating this, we have $u'(x)v(x)+u(x)v'(x)$. This means that

$$\int \left[u'(x)v(x) + u(x)v'(x)\right]\mathrm{d}x = u(x)v(x) + C$$

It is difficult to apply this formula directly, as we would need to recognize the function we are trying to integrate as having the form on the left. It is more useful to break up the left side:

$$\int u'(x)v(x)\mathrm{d}x + \int u(x)v'(x)\mathrm{d}x = u(x)v(x) + C$$

$$\rightsquigarrow \int u(x)v'(x)\mathrm{d}x = u(x)v(x) - \int u'(x)v(x)\mathrm{d}x$$

(we can absorb the $+C$ into the indefinite integral on the right). In other words, if F is an antiderivative of $u'(x)v(x)$, then $u(x)v(x) - F(x)$ is an antiderivative of $u(x)v'(x)$, as can be checked using the product rule. Applying this formula is called *integration by parts*; it in effect allows us to reduce the computation of an integral of a product to the integral of different product, moving the derivative from one factor to the other.

As with substitution, there is a concise notation for this. Writing $\mathrm{d}u = u'(x)\mathrm{d}x$ and $\mathrm{d}v = v'(x)\mathrm{d}x$, the formula can be written

$$\int u\mathrm{d}v = uv - \int v\mathrm{d}u$$

This method of integration is useful when differentiating one factor in a product makes it simpler. A particular case where this is often useful is when one of the factors is a power of x.

Example 10. Consider the integral

$$\int x \sin(x) \mathrm{d}x$$

To evaluate this, we will take $u = x$, and v so that $\mathrm{d}v = \sin(x)\mathrm{d}x$. In other words, v should be an antiderivative of $\sin(x)$, so we can choose $v = -\cos(x)$. We also have $\mathrm{d}u = \mathrm{d}x$. [Write table of $u, v, \mathrm{d}u, \mathrm{d}v$]. The integration by parts formula then reads

$$\int x \sin(x) \mathrm{d}x = -x\cos(x) - \int [-\cos(x)]\, \mathrm{d}x$$

This comes out to

$$-x\cos(x) + \int \cos(x)\mathrm{d}x = \sin(x) - x\cos(x) + C$$

As with substitution, we may need to apply integration by parts more than once.

Example 11. Consider

$$\int x^2 \exp(x) \mathrm{d}x$$

Taking $u_1 = x^2$, $\mathrm{d}v_1 = \exp(x)\mathrm{d}x$, we can take $v_1 = \exp(x)$, and obtain

$$x^2 \exp(x) - \int 2x \exp(x) \mathrm{d}x$$

Now taking $u_2 = 2x$, $\mathrm{d}v_2 = \exp(x)\mathrm{d}x$, $v_2 = \exp(x)$, we obtain

$$x^2 \exp(x) - \left[2x\exp(x) - \int 2\exp(x)\mathrm{d}x\right] = x^2\exp(x) - 2x\exp(x) + 2\exp(x) + C$$

Using the fundamental theorem of calculus, we can also apply integration by parts to definite integrals, using the formula

$$\int_a^b u(x)v'(x)\mathrm{d}x = [u(x)v(x)]\Big|_a^b - \int_a^b u'(x)v(x)\mathrm{d}x$$

Example 12. Consider the integral

$$\int_1^2 x^4 \ln(x) \mathrm{d}x$$

Taking $u = \ln(x)$, $\mathrm{d}v = x^4\mathrm{d}x$, we have $\mathrm{d}u = x^{-1}\mathrm{d}x$, and we can take $v = x^5/5$. Our integral then becomes

$$\left[\frac{\ln(x)x^5}{5}\right]\Big|_1^2 - \int_1^2 \frac{x^4}{5}\mathrm{d}x = \frac{32\ln(2)}{5} - \left[\frac{x^5}{25}\right]\Big|_1^2 = \frac{32\ln(2)}{5} - \frac{31}{25}$$

Unlike the case of substitution, the formula for definite integrals does not save much effort; applying the indefinite integral version and then only introducing the bounds at the end may even be easier.

It takes practice to identify situations where integration by parts can be usefully applied. There will be a lot of examples for you to practice on the homework.

We can sometime apply integration by parts even in situations where the function does not appear to be a product, by taking $v = x$, so $\mathrm{d}v = \mathrm{d}x$. We then obtain the general formula

$$\int u(x)\mathrm{d}x = x \cdot u(x) - \int x \cdot u'(x)\mathrm{d}x$$

Try applying this to a power function x^n and see what happens.

Example 13. We will use this method to find an antiderivative of the natural logarithm. Taking $u(x) = \ln(x)$ in the above formula, we obtain

$$\int \ln(x)\mathrm{d}x = x\ln(x) - \int x\frac{1}{x}\mathrm{d}x = x\ln(x) - \int 1\mathrm{d}x = x\ln(x) - x + C$$

Because a general logarithm function $\log_a(x)$ is defined to be $\ln(x)/\ln(a)$, a constant multiple of $\ln(x)$, we have the following conclusion.

Proposition. *For $a > 0$, $a \neq 1$, we have*

$$\log_a(x) = x\log_a(x) - \frac{x}{\ln(a)} + C$$

On the homework I will ask you to use a similar method to find antiderivatives of inverse trigonometric function.

Integration by parts essentially reduces the problem of computing one integral:

$$\int u(x)v'(x)\mathrm{d}x$$

to the problem of computing another integral

$$\int u'(x)v(x)\mathrm{d}x$$

As we have seen above, this is useful when the latter is more simple than the former. It can be useful, however, even if it does not make the integral simpler, in the case where applying it (multiple times) takes us back to the integral we started with. Let me illustrate.

Example 14. Consider the indefinite integral

$$\int \exp(x)\sin(x)\mathrm{d}x$$

We will apply integration by parts first with $u_1 = \sin(x)$, $v_1 = \exp(x)$, so $\mathrm{d}u_1 = \cos(x)\mathrm{d}x$, $\mathrm{d}v_1 = \exp(x)\mathrm{d}x$. Our integral becomes

$$\sin(x)\exp(x) - \int \cos(x)\exp(x)\mathrm{d}x$$

The integral we now have to evaluate is similar to the one we started with. Let's apply integration by parts again, with $u_2 = \cos(x)$, $v_2 = \exp(x)$, so $\mathrm{d}u_1 = -\sin(x)\mathrm{d}x$, and $\mathrm{d}v_2 = \exp(x)\cos(x)$. Our integral then becomes

$$\begin{aligned}\sin(x)\exp(x) - \left[\cos(x)\exp(x) - \int [-\sin(x)\exp(x)]\,\mathrm{d}x\right] \\ = \sin(x)\exp(x) - \cos(x)\exp(x) - \int \sin(x)\exp(x)\mathrm{d}x\end{aligned}$$

Now we are back to where we started, but we have gained information. Letting I represent the integral we want, we have derived the equation

$$I = (\sin(x) - \cos(x))\exp(x) - I$$

We can solve for I in this equation, as follows:

$$2I = (\sin(x) - \cos(x))\exp(x) + C \rightsquigarrow I = \frac{(\sin(x) - \cos(x))\exp(x))}{2} + C$$

That is,

$$\int \exp(x)\sin(x)\mathrm{d}x = \frac{(\sin(x) - \cos(x))\exp(x)}{2} + C$$

You can check that this is right by differentiating.